ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

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Many-Body Scattering, Energy Relaxation and Advanced Quantum Devices

I. Moving Beyond Coherent Transport

II. Phenomenological Models and Inelastic Mechanics

III. Advanced Quantum Device Applications

I. Moving Beyond Coherent Transport

♦ **Limitations of ideal case: Transport was assumed to be strictly coherent (elastic scattering only; electrons never lose energy inside the device)**

♦ **The Reality of Nanodevices: Electrons collide with vibrations (phonons), photons, and other electrons, causing heat generation and phase decoherence**

♦ **Goal of this lecture: Expand the NEGF formalism to handle non-elastic scattering via Inelastic Self-Energies, and apply it to cutting-edge quantum devices**

Previously, electrons have been treated as coherent quantum waves traversing a device without energy loss to the environment. Although this approximation works well for ultra-short channels, it breaks down in real-world devices where electron–phonon scattering induces electrical resistance and heating. The present discussion moves beyond the coherent limit, introducing the mathematical incorporation of energy relaxation and phase decoherence into NEGF, and exploring how this framework explains the operation of advanced quantum devices

I. Moving Beyond Coherent Transport

Real-Space Collisions — Decoherence and Energy Relaxation:

Two critical effects of real-world scattering:

- **Phase Decoherence (Elastic/Inelastic): Destroys the pure phase relationship of the electron wave, damping out quantum interference loops**
- **Energy Relaxation (Inelastic): Electrons emit or absorb energy quanta, shifting their position along the energy axis ($E \rightarrow E \pm \hbar\omega$)**

Why the Landauer formula fails here: $T(E)$ assumes an electron enters at energy E and must exit at exactly energy E

An electron traveling through a real crystal lattice undergoes two types of scattering. The first is phase decoherence, which scrambles the quantum wave phase and washes out fine quantum interference patterns. The second is energy relaxation, wherein a phonon (a localized heat packet) is emitted and the electron drops to a lower energy state. Since the Landauer formula requires identical injection and ejection energies, it fails in this regime. NEGF overcomes this limitation by dynamically coupling different energy states together

I. Moving Beyond Coherent Transport

Generalizing the Self Energy Concept to Many-Body Interactions:

♦ **Lead self-energies (Σ_L, Σ_R) represent the escape of electrons into open spatial boundaries**

♦ **Conceptual Leap: We introduce a Scattering Self Energy ($\Sigma_{scattering}$) that represents the 'escape' of electrons into other internal degrees of freedom (like the phonon bath)**

♦ **Total Retarded Self Energy of the system now becomes an additive operator:** $\Sigma^R_{total}(E) = \Sigma^R_L(E) + \Sigma^R_R(E) + \Sigma^R_{scatteromg}(E)$

The exact same mathematical concept is used to handle scattering. A 'Scattering Self Energy' is introduced. Instead of escaping into a physical lead, this term describes the electron escaping into a different energy state or shifting its quantum phase. By simply adding this scattering term to the total self-energy, the mathematical structure of the Green's functions remains completely intact

I. Moving Beyond Coherent Transport

How Inelastic Scattering Reshapes Available States

The updated central Retarded Green's function containing scattering: $G_C^R(E) = \left[EI - H_C - \Sigma_L^R(E) - \Sigma_R^R(E) - \Sigma_{scattering}^R(E)\right]^{-1}$

Physical Interpretation of $Imag\left[\Sigma_{scattering}^R(E)\right]$**: Defines the intrinsic scattering rate,** $\tau_{scattering}^{-1}(E)$ **and adds lifetime broadening to the spatial electronic states**

The spectral function $A(E)$ **spreads out further, shifting and dampening localized resonant states**

The presence of scattering modifies the Retarded Green's Function, $G_C^R(E)$**, through the scattering self-energy entering the denominator directly. The imaginary part of this scattering term represents the physical scattering rate, inversely proportional to the particle's quantum lifetime. This lifetime broadening dampens and smears out the electronic states. Sharp resonant peaks that allowed perfect tunneling in a coherent model become wider and shorter, capturing the real-world loss of quantum resonance due to collisions**

I. Moving Beyond Coherent Transport

Tracking Inelastic State Population:

To find the non-equilibrium electron population, we must also update the Keldysh relation using the Lesser Scattering Self-Energy ($\Sigma^{<}_{scattering}$): $G^{<}(E) = G^{R}(E)\left[\Sigma^{<}_{L}(E) + \Sigma^{<}_{R}(E) + \Sigma^{<}_{scattering}(E)\right]G^{A}(E)$

Physical Meaning:

- $\Sigma^{<}_{L}$, $\Sigma^{<}_{R}$ — **Describe electrons being injected into the channel from the left and right physical leads**
- $\Sigma^{<}_{scattering}$: **Describes electrons being 'injected' into energy state E from other energy states via scattering transitions**

To find out how electrons populate these states out of equilibrium, the Keldysh equation is revisited. A new term now appears inside the brackets: the Lesser Scattering Self Energy, $\Sigma^{<}_{scattering}$. While the lead terms represent electrons physically entering from the outside world, this new scattering term represents electrons being 'scattered into' this specific energy level from higher or lower energies. It acts as an internal, kinetic injection mechanism

I. Moving Beyond Coherent Transport

The Interlocked Loop of Quantum Scattering:

The Fundamental Problem: The scattering self energy $\Sigma^{<}_{scattering}$ is a direct function of the Green's function itself $\Sigma^{<}_{scattering} = \mathcal{F}(G)$

This requires the Self-Consistent Born Approximation (SCBA)

Knowledge of G is required to determine electron positions for calculating the scattering probability $\Sigma^{<}_{scattering}$, yet $\Sigma^{<}_{scattering}$ is itself required to compute the updated states in G

A theoretical circle arises at this point, known as the Self-Consistent Born Approximation (SCBA). Calculation of the scattering self-energy requires knowledge of the density of available electronic states, yet these states are themselves modified by the scattering process. Consequently, scattering cannot be solved in a single step. An iterative loop must be implemented, in which the Green's functions and scattering self-energies are successively updated until convergence to a stable equilibrium is achieved

II. Phenomenological Models and Inelastic Mechanics

Acoustic Phonon Scattering: The Deformation Potential Model

Acoustic phonons have very low energy ($\hbar\omega \approx 0$), meaning they scatter electrons elastically but destroy their quantum phase. Formulation within SCBA in the real-space representation:

$$\Sigma_{ac}^{R}(r,E) = D_{ac} \cdot G^{R}(r,r,E)$$

$$\Sigma_{ac}^{<}(r,E) = D_{ac} \cdot G^{<}(r,r,E)$$

D_{ac} — Coupling constant proportional to the material's acoustic deformation potential and temperature T

Acoustic phonon scattering serves as a concrete example. Because acoustic lattice vibrations carry very little energy, they are treated as an elastic process. The mathematical form within the SCBA shows the self-energy at a specific coordinate to be simply a constant multiplied by the Green's function at that same coordinate. This local coupling strongly degrades the phase of the electron wave as temperature increases, accurately capturing thermal resistance without any energy shift of the electron

II. Phenomenological Models and Inelastic Mechanics

Optical Phonon Scattering: Energy Interconversion

Optical phonons possess a fixed, discrete high energy ($\hbar\omega_0$). Collisions force the electron to jump across energies
Mathematical formulation for the inelastic injection term:

$$\Sigma^{<}_{opt}(r,E) = D_{opt}\left[N_q G^{<}(E-\hbar\omega_0) + \left(N_q+1\right)G^{<}(E+\hbar\omega_0)\right]$$

N_q **— Phonon population given by the Bose-Einstein distribution.**

First term: Absorption of a phonon (gaining energy); Second term: Emission of a phonon (losing energy/generating heat)

Interactions between electrons and high-energy optical phonons introduce strictly inelastic physics. In the equation for $\Sigma^{<}_{opt}$, the self energy at energy E depends on the Green's functions at $E-\hbar\omega_0$ and $E+\hbar\omega_0$. This dependence provides the mathematical proof of energy relaxation: an electron at higher energy emits an optical phonon, drops by $\hbar\omega_0$, and is injected into a lower-energy channel, thereby heating the device lattice

II. Phenomenological Models and Inelastic Mechanics

The Büttiker Virtual Probe: Simplification of Complex Scattering

Calculating full microscopic SCBA equations is computationally demanding

The Büttiker Probe Concept: Attach a virtual, fictitious third lead to a specific point inside the device channel

The virtual lead pulls an electron out of the coherent channel, completely scrambles its phase and energy inside a dummy reservoir, and then reinjects it back into the device

Net current condition: To ensure charge conservation, the net current flowing into the virtual probe must be locked to exactly zero ($I_{probe} = 0$)

Evaluating full electron-phonon equations can be incredibly time-consuming. To capture the core physics of scattering without the massive mathematical overhead, Markus Büttiker introduced a brilliant conceptual tool: the Virtual Probe. In this approach, a microscopic, imaginary wire is attached to the center of the channel. This wire extracts an electron, completely scrambles its quantum phase inside a fictitious reservoir, and reinjects it. By enforcing a rule that this virtual wire cannot store or leak any net charge, the effects of real physical scattering are perfectly mimicked with a fraction of the computational effort

II. Phenomenological Models and Inelastic Mechanics

The Current Continuity Equation Under Scattering:

Under inelastic scattering, current is no longer constant at each individual energy slice. Electrons shift between energy lanes as they flow down the channel

The total current must be evaluated by summing up the spatial divergence of the Keldysh system:

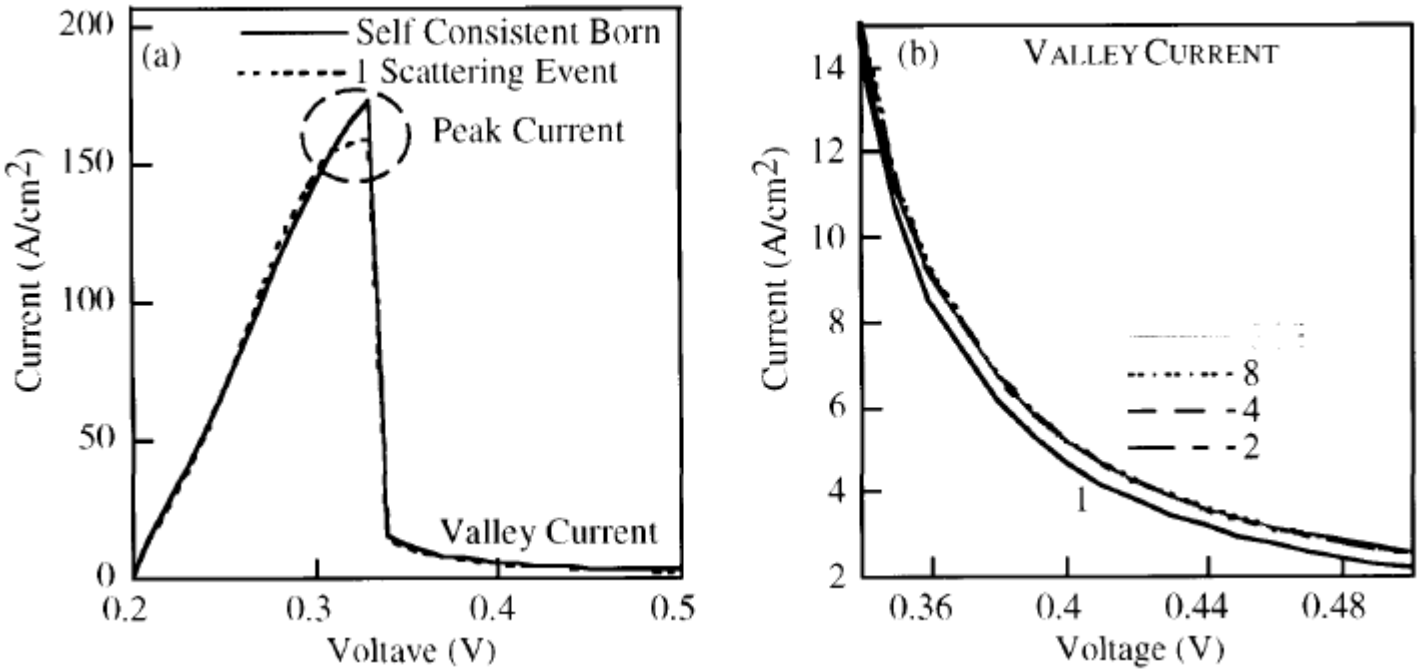
$$I = \frac{2e}{h}\int_{-\infty}^{\infty} Trace[\Sigma^{<}(E)G^{>}(E) - \Sigma^{>}(E)G^{<}(E)]dE$$

This equation proves that local current conservation is strictly maintained across the entire real-space grid, even as electrons rearrange themselves energetically

In a coherent device, the current flowing at energy *E* remains constant from source to drain. Under inelastic scattering, this constancy breaks down, as electrons change energy lanes mid-journey. Calculation of the total current requires integration of a broader equation that tracks the balance between electrons entering and leaving each energy slice. This generalized formula guarantees exact conservation of total electrical current at every cross-section of the device, despite inter-state energy shifts

III. Advanced Quantum Device Applications

Application 1: Resonant Tunneling Diodes



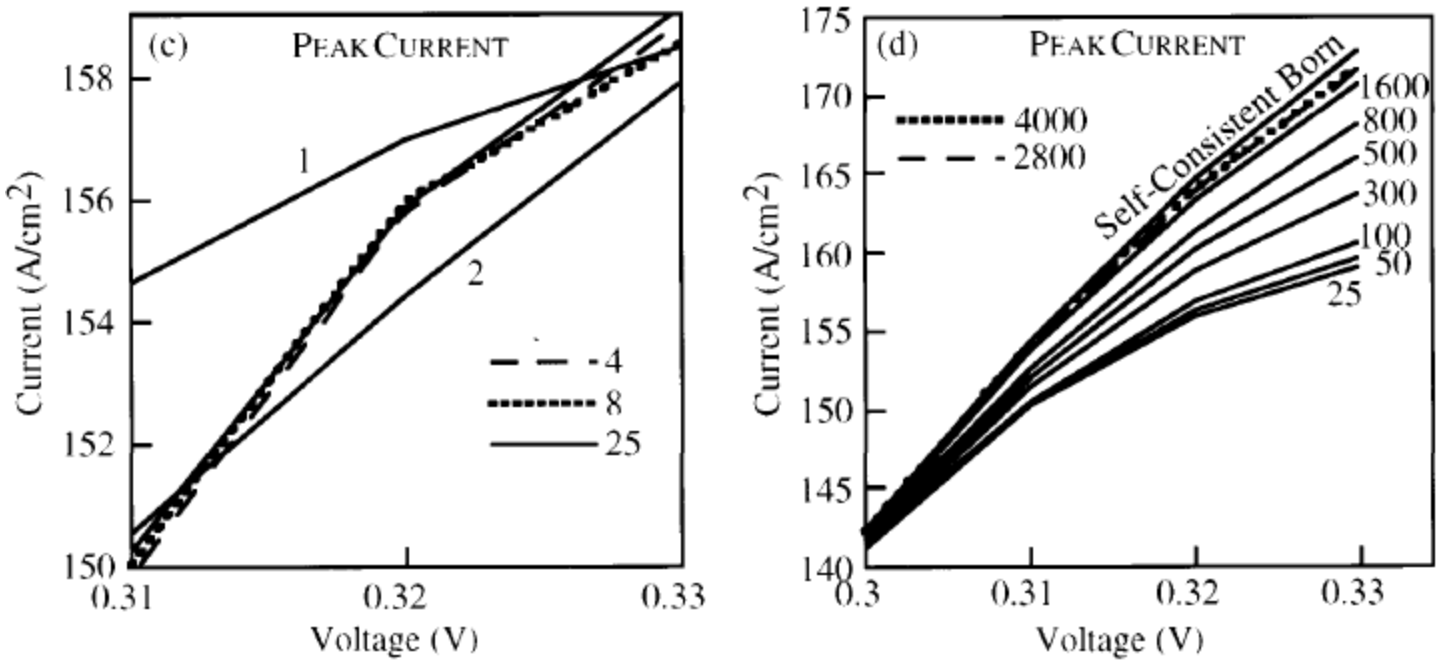
Numerical demonstration of the convergence of the multiple sequential scattering algorithm and the self-consistent Born algorithm. (*a*) I-V of GaAs/AlAs RTD calculated using the self-consistent Born algorithm and the MSS algorithm with one scattering event. Convergence of the MSS algorithm to the SCB algorithm: (*b*) in the valley current region

- **Device Physics: A GaAs/AlGaAs quantum well sandwiched between two potential barriers creating discrete, quasi-bound states**
- **The Coherent Overestimation: Purely coherent models ($T(E)$ under Landauer) predict an infinitely sharp transmission peak and a valley current that drops to near zero**
- **The Scattering Reality: Inelastic scattering self energy ($\Sigma_{scattering}$) broadens the localized state lifetime, smearing the transmission spectrum and generating a parasitic "valley current" that degrades the Peak-to-Valley Ratio (PVR)**

Roger Lake, Gerhard Klimeck, R. Chris Bowen and Dejan Jovanovic, Single and multiband modeling of quantum electron transport through layered semiconductor devices, *J. Appl. Phys.* 81 (12), 15 June 1997

III. Advanced Quantum Device Applications

Application 1: Resonant Tunneling Diodes



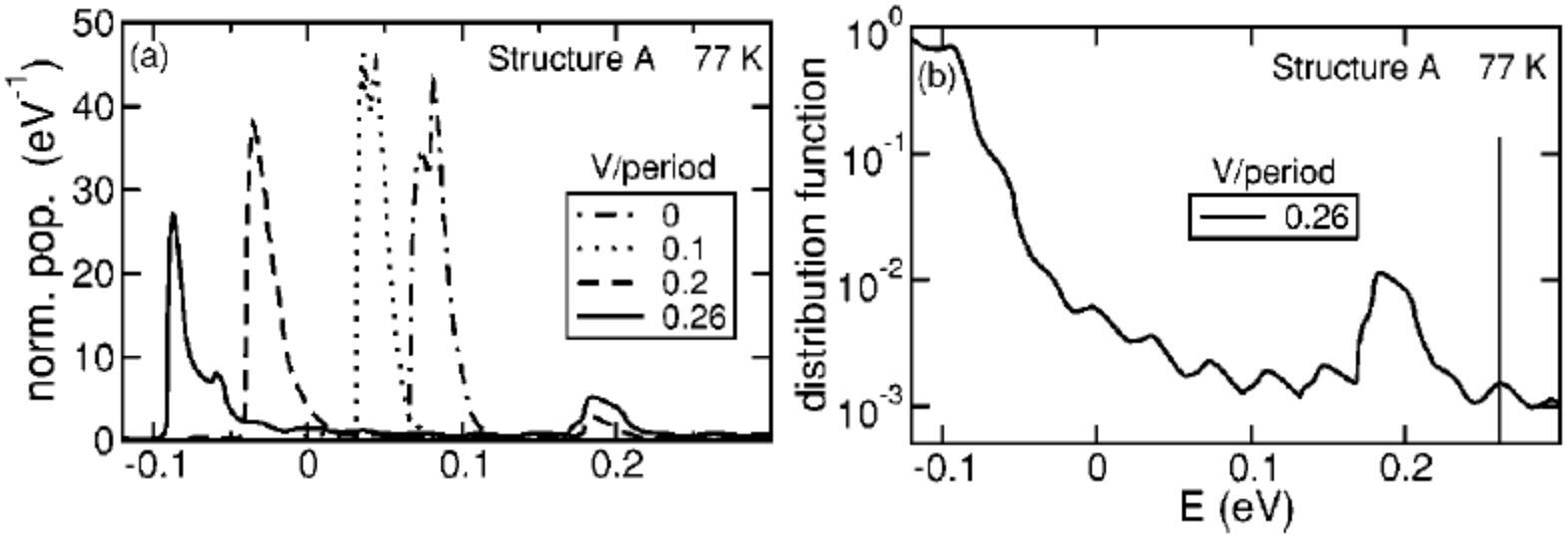
(*c*) in the peak current region for few scattering events, N51225, and (*d*) in the peak current region for N=25-4000

The Resonant Tunneling Diode (RTD) serves as the first real-world application. In a landmark paper by Lake and Klimeck, the *I-V* curve of an RTD was plotted under two different assumptions. The dashed line corresponds to the strictly coherent model, where the current drops sharply to almost zero in the valley, yielding an unrealistically high peak-to-valley ratio. With scattering self-energies included (solid line), electron–phonon interactions broaden the resonant state inside the quantum well via a finite lifetime (Imag[$\Sigma_{scattering}$]). This broadening smears the transmission peak and produces a significant background valley current, in excellent agreement with experimental measurements

Roger Lake, Gerhard Klimeck, R. Chris Bowen and Dejan Jovanovic, Single and multiband modeling of quantum electron transport through layered semiconductor devices, *J. Appl. Phys.* 81 (12), 15 June 1997

III. Advanced Quantum Device Applications

Application 2: Quantum Cascade Lasers



(a) $n(E)$, normalized population per period for different applied voltages. (b) $f(E)$, distribution function in one period at 0.26 V/period. The thin vertical line marks the position of the conduction-band offset between the thin (first) well in the active region in this period, and the thick barrier separating this well from the preceding period. The bottom of this well is at around 0 eV. The peak at around 0.2 eV corresponds to a population inversion in the upper laser level in the active region

♦ **Device Physics: Unipolar semiconductor lasers where electrons cascade down an engineered atomic staircase, emitting a photon at each step**

♦ **The Engineering Challenge: Continuous laser action requires maintaining a strict population inversion between the upper and lower laser states ($n_{upper} > n_{lower}$)**

S.-C. Lee and A. Wacker, Nonequilibrium Green's function theory for transport and gain properties of quantum cascade structures, *Phys. Rev. B* 66, 245314 (2002)

III. Advanced Quantum Device Applications

Application 2: Quantum Cascade Lasers

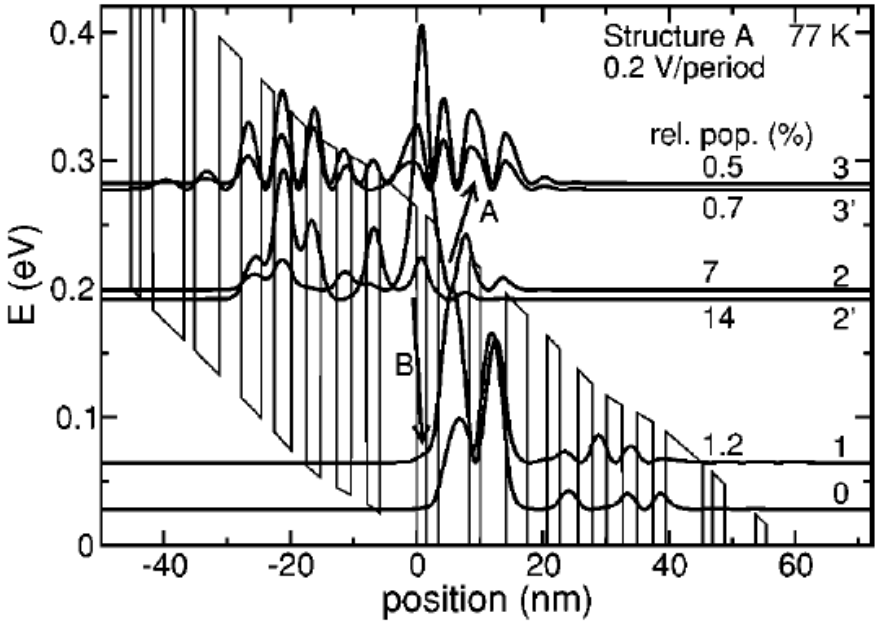


FIG. 13. Wannier-Stark levels in structure *A* at 0.2 V/period.

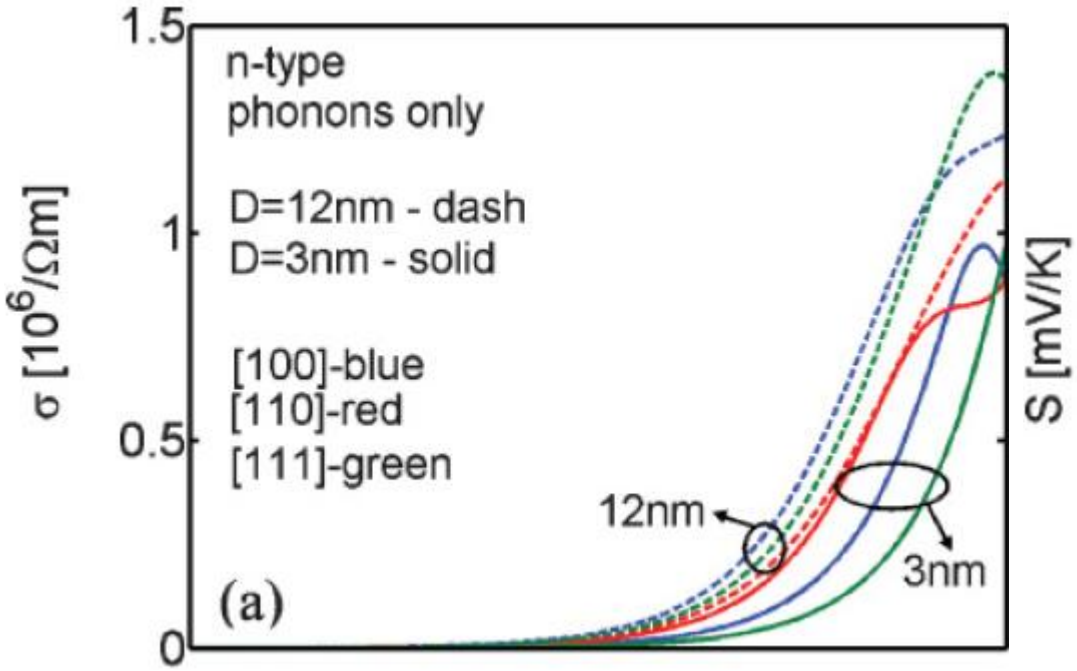
♦ **NEGF Design Solution: Engineers use NEGF to deliberately tune the energy spacing of the lower subbands to match the longitudinal optical (LO) phonon energy ($\Delta E \approx 36$ meV for GaAs). This forces $\Sigma^{<}_{opt}$ to maximize, emptying the lower state in less than 1 picosecond**

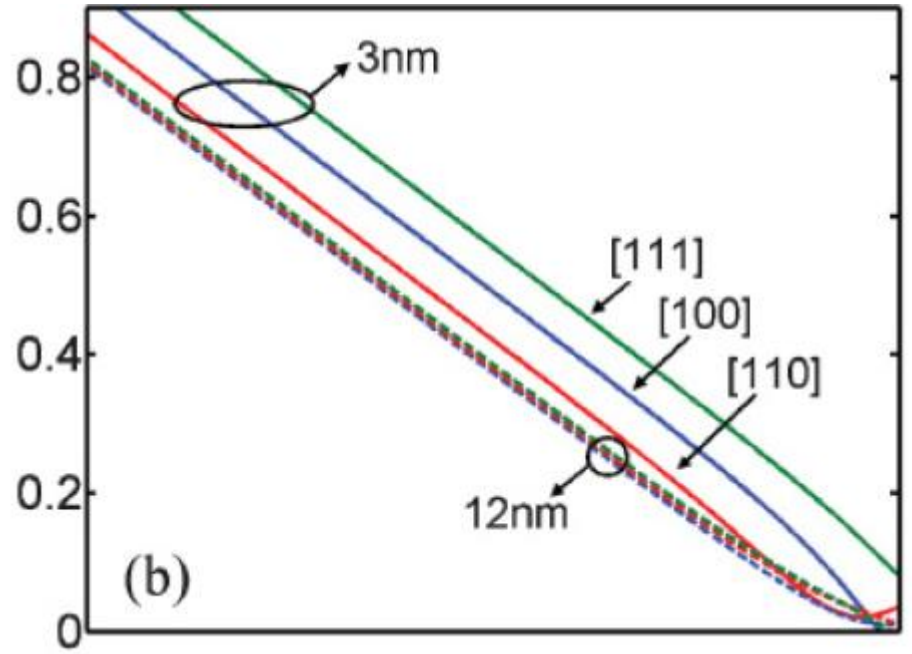
Quantum Cascade Lasers represent a triumph of quantum transport engineering. Maintaining QCL operation requires the lower laser level to be emptied almost instantly to preserve population inversion. In Lee and Wacker's PRB paper, inelastic NEGF was employed to compute the exact electron population ($G^{<}$) in each subband. Atomic layer thicknesses were designed so that the lower state spacing precisely matches the optical phonon energy of 36 meV, bringing the scattering self-energy term $\Sigma^{<}_{opt}$ into resonance. NEGF demonstrates that this resonance forces electrons to dissipate energy as heat via sub-picosecond phonon emission, thereby clearing the lower state and sustaining laser activity

S.-C. Lee and A. Wacker, Nonequilibrium Green's function theory for transport and gain properties of quantum cascade structures, *Phys. Rev. B* 66, 245314 (2002)

III. Advanced Quantum Device Applications

Application 3: Thermoelectric Transport and Energy Filtering





(Color online) TE coefficients for *n*-type NWs in [100] (blue), [110] (red), and [111] (green) transport orientations for diameters D = 3 nm (solid lines) and D = 12 nm (dashed lines) vs carrier concentration. (*a*) The electrical conductivity. (*b*) The Seebeck coefficient

♦ **Physics Goal: Maximize the thermoelectric figure of merit $ZT = {S^2\sigma T}/{\kappa}$. This requires blocking 'cold' carriers while allowing 'hot' carriers to pass**

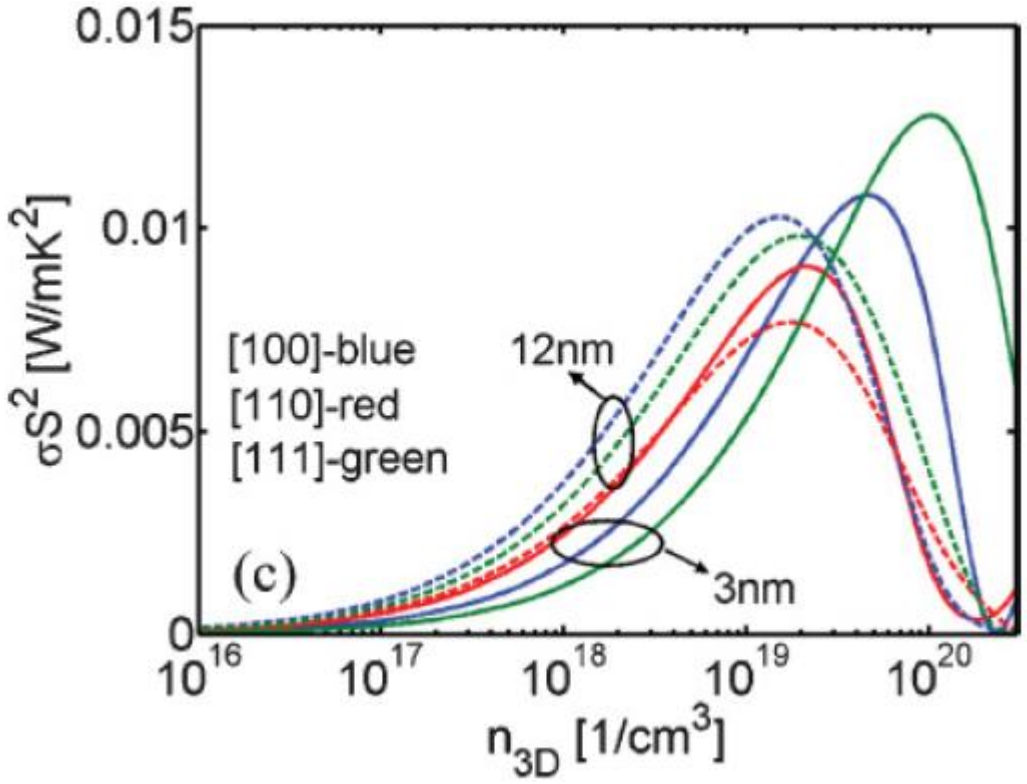
♦ **The Energy Filtering Concept: Introduce nanoscale potential barriers or embedded impurities that act as a high-pass filter for the electronic stream**

♦ **NEGF Quantum Quantifier: NEGF explicitly computes how these nanostructures chop off the low-energy tail of the transmission function *T*(*E*), allowing direct, non-equilibrium optimization of the Seebeck coefficient (*S*)**

Neophytos Neophytou and Hans Kosina, Effects of confinement and orientation on the thermoelectric power factor of silicon nanowires, *Phys. Rev. B* 83, 245305 (2011)

III. Advanced Quantum Device Applications

Application 3: Thermoelectric Transport and Energy Filtering



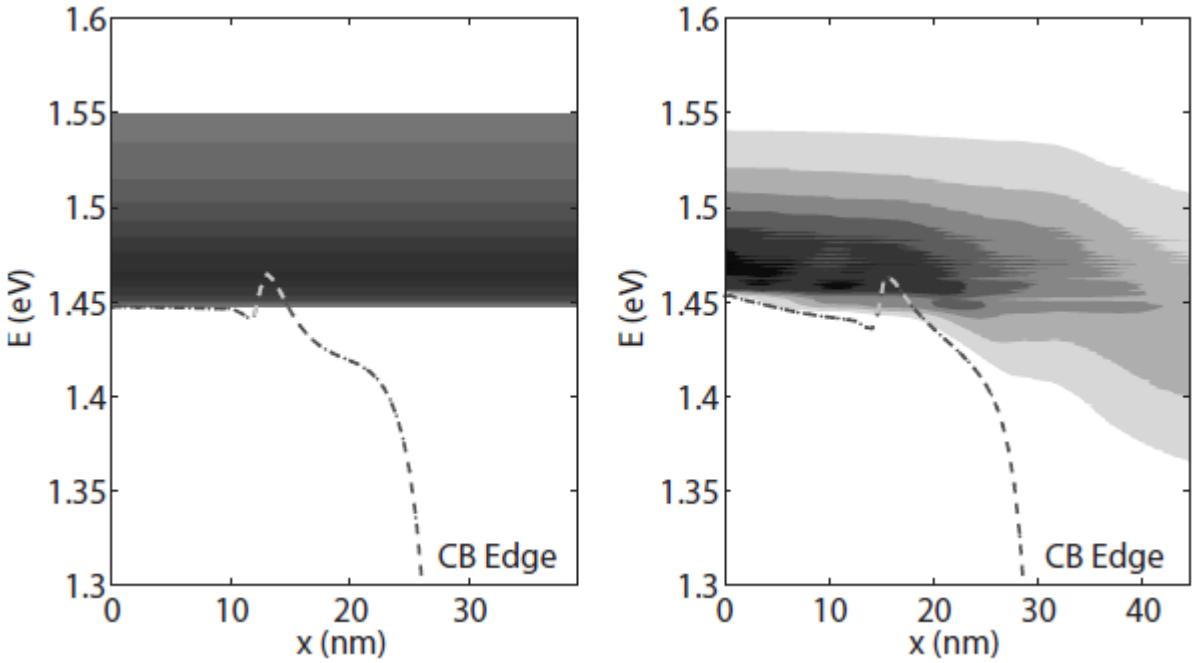
(*c*) The power factor. Only phonon scattering (ADP and IVS) is considered

In the work of Neophytou and Kosina, transport through silicon nanowires was simulated. The transmission spectrum *T*(*E*) on the left shows transmission pinned to zero at low energies, as nanoscale barriers completely block or scatter low-energy 'cold' electrons. Only high-energy 'hot' electrons possess the transmission states required for passage. NEGF provides the precise mathematical framework to shape the *T*(*E*) profile, enabling the Seebeck coefficient to be maximized without compromising overall electrical conductivity

Neophytos Neophytou and Hans Kosina, Effects of confinement and orientation on the thermoelectric power factor of silicon nanowires, *Phys. Rev. B* 83, 245305 (2011)

III. Advanced Quantum Device Applications

Application 4: Nanoelectronic Scaling and Subthreshold Leakage



Spectral current of the Si GAA NW FET with a diameter d = 2.5 nm from Fig. 4 at V_{gs} = 0.5 V and V_{ds} = 0.6 V. The left subplot represents the ballistic current homogeneous and the right subplot the current with electron-phonon scattering. The dashed lines indicate the position of the CB edge

- **The Classical Limit: Conventional Field-Effect Transistors (FETs) are limited by the thermal energy tail of injecting electrons, dictating a minimum Subthreshold Swing (*SS*) of 60 mV/decade at room temperature**
- **The Sub-10nm Quantum Crisis: As the physical gate length drops below 10 nm, classical thermionic models completely fail**
- **The NEGF Revelation: Open-boundary NEGF calculations reveal a massive, parasitic current component: direct source-to-drain quantum tunneling through the potential barrier when the device is supposedly turned off**

Mathieu Luisier and Gerhard Klimeck, Atomistic full band simulations of silicon nanowire transistors Effects of electron phonon scattering, *Phys. Rev. B* 80, 155430 (2009)

III. Advanced Quantum Device Applications

Application 5: Tunnel Field-Effect Transistors

♦ **The TFET Concept: Replace thermal injection with a gated Band-to-Band Tunneling (BTBT) mechanism using a *p-i-n* semiconductor architecture**

♦ **The Natural Energy Filter: The valence band edge of the *p* type source cuts off the high-energy thermal tail of electrons, enabling an ultra-sharp turn-on response (SS < 60 mV/dec)**

♦ **Multi-Band NEGF Requirement: Capturing this requires an advanced multi-band tight-binding basis (like $sp^3d^5s^*$) to track the quantum mechanical overlap between valence and conduction band wavefunctions**

Scaling transistors down to the atomic limit encounters a physical wall known as the Boltzmann Tyranny, which limits classical switching to 60 mV per decade. Professor Luisier's atomistic NEGF simulations reveal that the reality below 10 nm is even more severe. The spectral current heatmap shows the energy-position distribution of current flow. Even with the gate barrier raised high to turn the transistor 'OFF', a bright horizontal stream passes directly through the center of the potential wall. This phenomenon is direct source-to-drain quantum tunneling, and NEGF is uniquely capable of tracking such wave-induced leakage

To overcome the 60 mV/dec limit, the Tunnel FET was designed. Rather than forcing electrons to hop over a barrier, they are allowed to tunnel horizontally from the valence band to the conduction band. On the right side of Luisier's paper, the I_d-V_g curve exhibits a steep slope that successfully breaks below the 60 mV/dec threshold. Modeling this requires more than a simple single-band Green's function. The Hamiltonian matrix must be extended to a multi-band tight-binding model, enabling the NEGF engine to simultaneously compute the quantum overlap between electron states in the conduction band and hole states in the valence band during their tunneling jump

Mathieu Luisier and Gerhard Klimeck, Atomistic full band simulations of silicon nanowire transistors Effects of electron phonon scattering, *Phys. Rev. B* 80, 155430 (2009)

III. Advanced Quantum Device Applications

Application 6: Optoelectronic NEGF

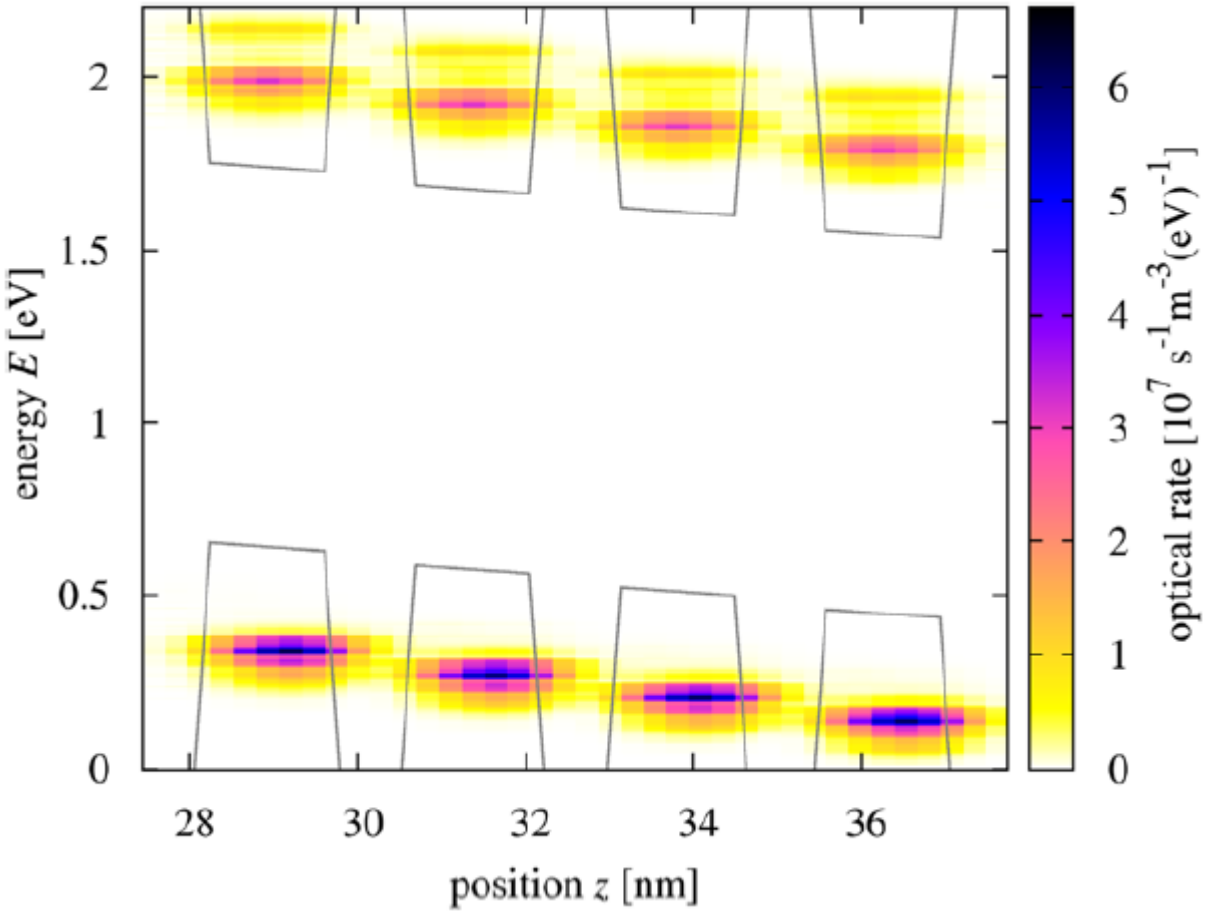


Fig. 8 Spatially and energy resolved charge carrier photogeneration rate in the quantum well region at short circuit conditions and under monochromatic illumination with energy $E_\gamma = 1.65$ eV and intensity $I_\gamma = 10\ \mathrm{kW/m^2}$ [105]

U. Aeberhard, Theory and simulation of quantum photovoltaic devices based on the non-equilibrium Green's function formalism, *J. Comput. Electron*. (2011) 10: 394-413

III. Advanced Quantum Device Applications

Application 6: Optoelectronic NEGF

♦ The Physics: Light-matter interaction requires introducing an electron-photon scattering self-energy matrix (Σ_{photon})

♦ The Quantum Lifecycle: A photon is absorbed, lifting an electron from the valence band to the conduction band, leaving a hole behind

♦ The NEGF Advantage: Traditional models separate optics and electronics into two separate software packages. NEGF unifies them: it models the optical generation of the carrier and its subsequent quantum transport to the open contacts within a single matrix

Application of NEGF to solar cells or photodetectors involves the addition of an electron-photon interaction term, Σ_{photon}. In the figures from Aeberhard's comprehensive paper, the spatial generation spectrum under light is displayed on the left. The unique feature of NEGF in this context is that it not only calculates photon absorption rates, but also simultaneously models the propagation, phase decoherence, and extraction of the generated quantum waves through the open contacts. This provides a complete bridge between quantum optics and electronic device transport within a single, elegant calculation loop

U. Aeberhard, Theory and simulation of quantum photovoltaic devices based on the non-equilibrium Green's function formalism, *J. Comput. Electron*. (2011) 10: 394-413

III. Advanced Quantum Device Applications

Application 7: Spintronics

A diagram illustrating the expansion of a scalar lattice site into a 2×2 matrix containing Spin-Up (↑) and Spin-Down (↓) block elements

♦ Beyond Spin-Degeneracy: To model spintronic devices, one must unlock the electron's magnetic moment. Every individual scalar element in the Green's function matrix is expanded into a 2×2 spinor block

♦ The Spinor Matrix Architecture:

♦ Physical Meaning: Diagonal elements represent pure spin-up/down states. Off-diagonal elements ($H^{\uparrow\downarrow}$) represent spin-orbit coupling or non-collinear magnetic fields that physically flip the electron's spin direction

Extending the NEGF framework into the field of Spintronics requires accounting for the electron's spin angular momentum. In the matrix structure underlying the PRL paper by Theodonis and coworkers, each site in the real-space grid is no longer a single scalar but a 2×2 sub-matrix. The diagonal elements track the separate pathways of spin-up and spin-down electrons. The off-diagonal terms capture the essential physics—they represent spin-orbit interactions or local magnetic moments that exert a quantum torque, enabling the physical flipping of a spin-up electron into a spin-down state during propagation

Ioannis Theodonis, Nicholas Kioussis, Alan Kalitsov, *et al.*, Anomalous Bias Dependence of Spin Torque in Magnetic Tunnel Junctions, *Phys. Rev. B.* 97, 237205 (2006)

III. Advanced Quantum Device Applications

Application 7: Spintronics

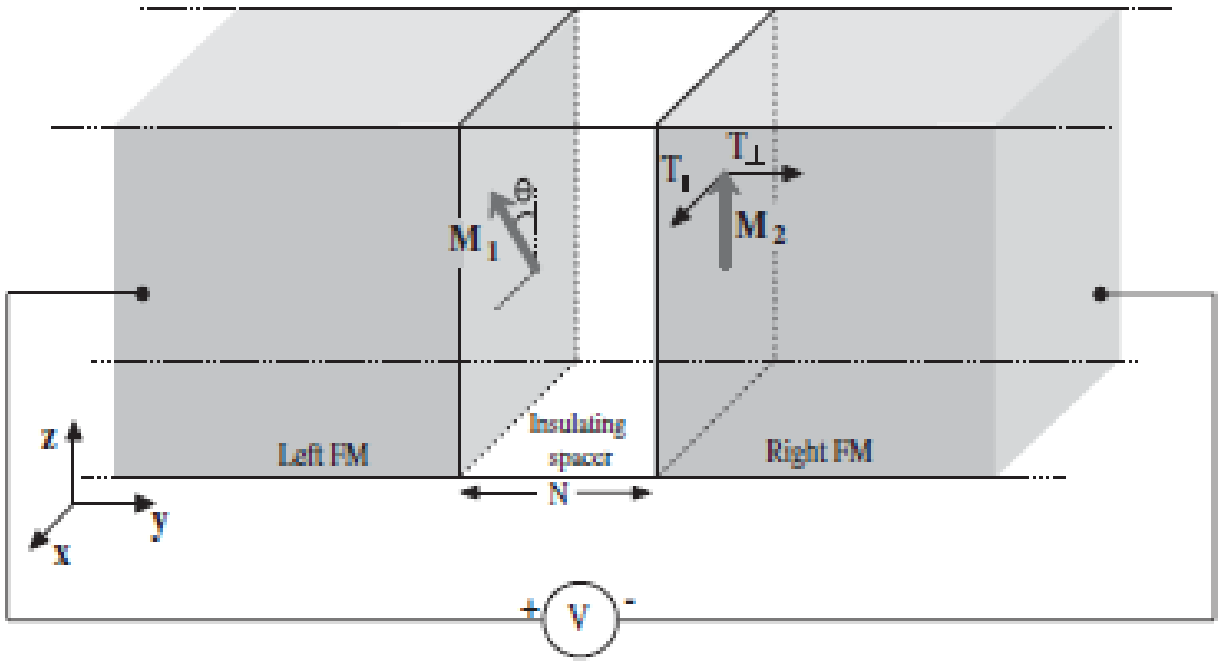


FIG. 1. Schematic structure of the MTJ, consisting of left and right semi-infinite FM leads separated by a thin nonmagnetic insulating system containing N atomic layers. The magnetization $\mathbf{M}_2$ of the right FM lead is along the z axis, whereas the magnetization $\mathbf{M}_1$ of the left lead is rotated by angle θ around the y axis with respect to $\mathbf{M}_2$.

Ioannis Theodonis, Nicholas Kioussis, Alan Kalitsov, *et al.*, Anomalous Bias Dependence of Spin Torque in Magnetic Tunnel Junctions, *Phys. Rev. B.* 97, 237205 (2006)

III. Advanced Quantum Device Applications

Application 8: Spin-Polarized Tunneling and Tunnel Magnetoresistance

Device Structure: Ferromagnet/Insulator/Ferromagnet thin-film stack (the fundamental memory bit of Spin-Transfer Torque MRAM)

The TMR Mechanism:

♦ Parallel (*P*) State: Injection spins align with available empty states in the receiving layer → High Transmission → Low Resistance (Logic '0')

♦ Anti-Parallel (AP) State: Injection spins face a deficit of matching states → Forbidden Tunneling → High Resistance (Logic '1')

NEGF Capability: Separately extracts $T_{\uparrow}(E)$ and $T_{\downarrow}(E)$ under high voltage biases to determine the reading margin of computer memory

Application of this spinor language to a Magnetic Tunnel Junction — the core element of modern MRAM chips — is now considered. The spin-resolved transmission plots from this PRL paper show that in the Parallel state, the spin-up transmission peak aligns perfectly with the available empty states, yielding a large current pathway. In the Anti-Parallel state, the transmission spectrum for both spins collapses near the Fermi level, as electrons encounter no available states of matching spin orientation when crossing the barrier. NEGF enables independent calculation of these dual transport lanes under high operating voltages, providing engineers with accurate reading margins for memory chips

Ioannis Theodonis, Nicholas Kioussis, Alan Kalitsov, *et al.*, Anomalous Bias Dependence of Spin Torque in Magnetic Tunnel Junctions, *Phys. Rev. B.* 97, 237205 (2006)

III. Advanced Quantum Device Applications

Application 8: Spin-Polarized Tunneling and Tunnel Magnetoresistance

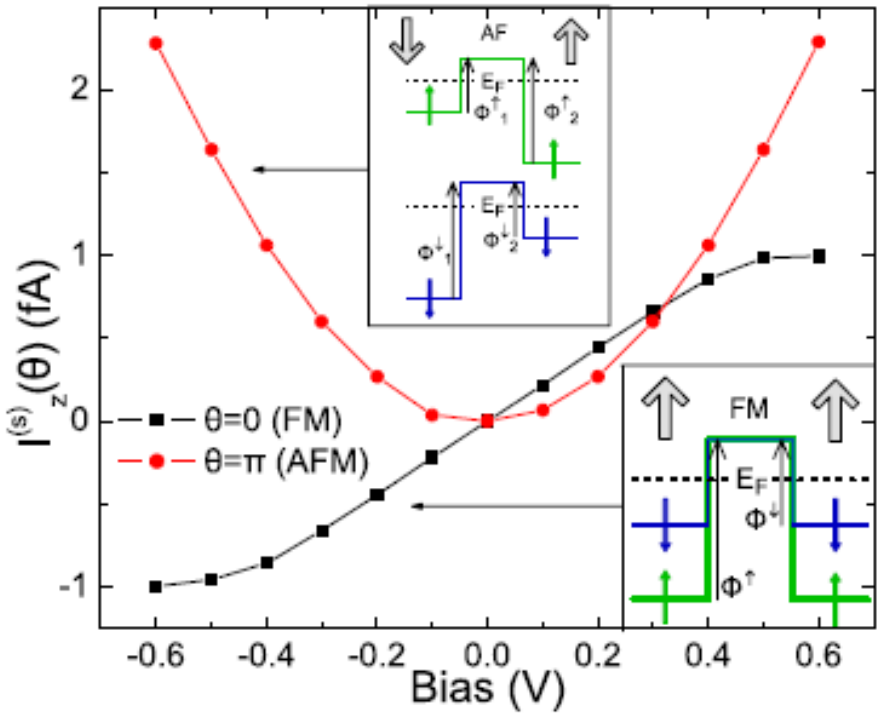


FIG. 5 (color online). Bias dependence of the spin-current density, $I_z^{(s)}(\theta) \equiv I^{\uparrow}(\theta) - I^{\downarrow}(\theta)$, for the FM and AF orientations, respectively. In the FM case the majority and minority electrons tunnel through a symmetric barrier (lower inset) with different barrier heights, Φ^{σ}, from the bottom of the band. In the AF case (upper inset), the two spin channels tunnel through asymmetric barriers with the same average barrier height, $\bar{\Phi}^{\sigma}$, but with a barrier asymmetry, $\Delta\Phi^{\sigma}$, of opposite sign.

Ioannis Theodonis, Nicholas Kioussis, Alan Kalitsov, *et al.*, Anomalous Bias Dependence of Spin Torque in Magnetic Tunnel Junctions, *Phys. Rev. B.* 97, 237205 (2006)

III. Advanced Quantum Device Applications

Application 8: Spin Transfer Torque (STT) Mechanics

♦ **The Writing Challenge: How to flip the magnetic orientation of an MTJ in order to write data without using power-hungry, bulky external magnetic coils?**

♦ **The STT Solution: Pass a highly spin-polarized current through the device. As electrons cross the interface, they transfer their spin angular momentum directly to the free layer's lattice**

♦ **The NEGF Torque Equation: The physical torque vector T is directly calculated by taking the mathematical trace of the Pauli spin matrices (σ) against the Keldysh lesser Green's function density matrix:**

$$T = \frac{1}{2\pi}\int Trace\{\sigma \cdot Im[G^{<}(E)]\} \times MdE$$

Data writing in MRAM without conventional magnetic coils is accomplished via Spin-Transfer Torque. A heavy current passed through the junction causes electrons to transfer their spin angular momentum to the material, generating a microscopic mechanical torque that reverses the magnetization. In the critical plot of Torque versus Bias Voltage from this PRL paper, the torque is seen to deviate from linearity—it oscillates and even changes sign at high biases. This complex non-equilibrium behavior is accessible only because the NEGF engine explicitly integrates the trace of the Pauli spin vector against the non-equilibrium density matrix $G^{<}$. This formula provides memory designers with exact information on the current pulse required to write a data bit

Ioannis Theodonis, Nicholas Kioussis, Alan Kalitsov, *et al.*, Anomalous Bias Dependence of Spin Torque in Magnetic Tunnel Junctions, *Phys. Rev. B.* 97, 237205 (2006)

III. Advanced Quantum Device Applications

Application 9: Topological Insulators (TIs) and Edge Transport

Symmetry Protected Helical Channels and Backscattering Immunity

The famous plot showing device Conductance (G) remaining perfectly locked on a flat, quantized plateau ($G = 1 \times \frac{2e^2}{h}$) even as the material disorder strength (W) increases drastically

♦ Materials Physics: Topological Insulators act as strict insulators in their interior bulk but support metallic, conducting states on their physical edges
♦ Helical Edge States: Spin-momentum locking mandates that spin-up electrons travel exclusively forward, while spin-down electrons travel exclusively backward
♦ The NEGF Proof of Robustness: Introduce a massive, highly disordered random potential scattering matrix ($\Sigma_{disorder}$) into the channel. NEGF demonstrates that the total transmission $T(E)$ remains completely unaltered

Topological Insulators are considered. These materials possess helical edge states where spin-up electrons move forward and spin-down electrons move backward. In a groundbreaking plot by Li and Shen, a massive disorder scattering self-energy ($\Sigma_{disorder}$) was introduced into their NEGF simulator. In a normal copper wire, such defect scattering would cause Anderson localization, dropping current to zero. However, the conductance remains flat at the quantized value of 11 over a wide range of disorder strength. NEGF proves that backscattering without spin-flip is forbidden, rendering the edge states immune to defects

Jian Li, Rui-Lin Chu, J. K. Jain and Shun-Qing Shen, Topological Anderson Insulator, *Phys. Rev. B.* 102, 136806 (2009)

III. Advanced Quantum Device Applications

Application 9: Topological Insulators (TIs) and Edge Transport

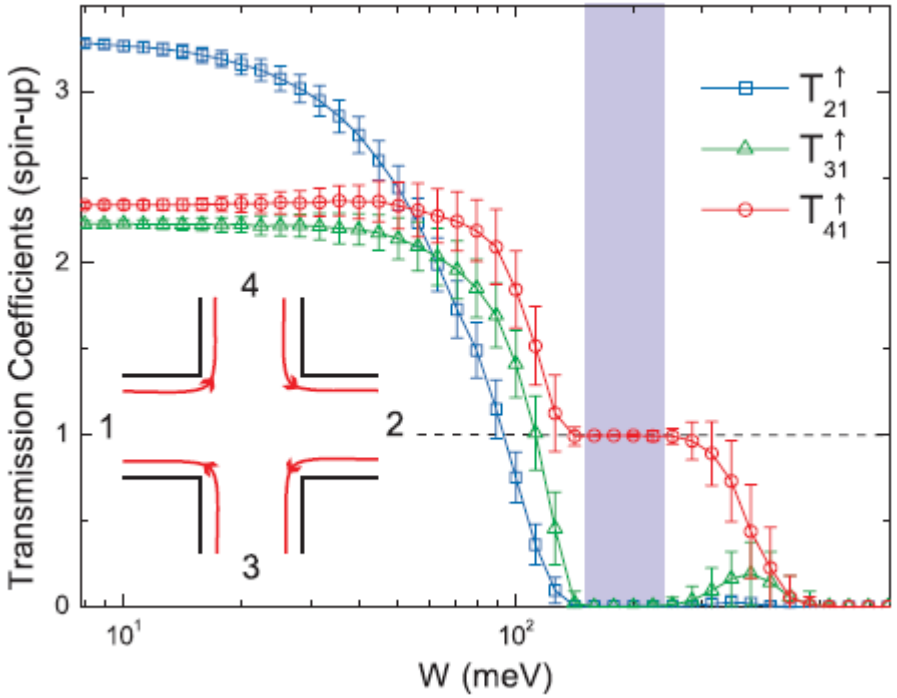


FIG. 4 (color). Three independent spin-resolved transmission coefficients, $T_{21}^{\uparrow}$, $T_{31}^{\uparrow}$, and $T_{41}^{\uparrow}$, are plotted as functions of disorder strength W. Standard deviations of the transmission coefficients for 1000 samples are shown as the error bars. In the shadowed range of disorder strength, all bulk states are localized and only chiral edge states exist, which is schematically shown in the inset (for spin-up component only). The width of leads is 500 nm and $M = 1$ meV and $E_f = 20$ meV.

Jian Li, Rui-Lin Chu, J. K. Jain and Shun-Qing Shen, Topological Anderson Insulator, *Phys. Rev. B.* 102, 136806 (2009)

III. Advanced Quantum Device Applications

Mapping the Boundaries of Quantum and Classical Transport

♦ Landauer-Büttiker Wave Formalism: Captures wave interference, phase coherence, and tunneling. However, it is restricted to elastic systems with simple boundary contacts

♦ Kubo Linear Response Formalism: Highly accurate for bulk, near-equilibrium material constants, but struggles under high voltage bias conditions and complex multi-terminal geometries

♦ The NEGF Engine: The comprehensive framework. Seamlessly unifies open spatial boundaries, arbitrary non-equilibrium bias voltages, and many-body inelastic scattering self-consistently

The complete hierarchy of transport physics can now be placed in perspective. Historically, classical models like the Drude and Boltzmann equations treated electrons as tiny billiard balls. The Landauer wave framework was then developed to capture quantum tunneling. While the Kubo method provides excellent insights into bulk material constants, NEGF stands out as the most comprehensive framework available, seamlessly unifying quantum wave mechanics, open boundaries, high voltage biases, and inelastic scattering into a single mathematical engine. The complete theoretical foundation of modern quantum transport has now been covered



That's all for this session

Thank you for your attention and great participation